Modern Academy
for Engineering and Technology in Maadi
Electrical Engineering

Academic Year	2025-2026
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Academic Level:	Senior 2

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCn425) Digital Signal Processing
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan
Branch: Communications & Computers
Time: 2 hours

Number of Pages: 1 | Number of Questions: 5 | Attempt all questions

Question 1 [8 Points] [LO's: 1]

a- In detail, Explain the quantization process used in quantization of continuous-amplitude signals. 2.5

b- For the following signal: $x_a(t) = 2 \cos(20000 \pi t)$,

(i) Plot this signal, $x_a(t)$. 1

(ii) Find its minimum sampling frequency. 1

(iii) Using the minimum sampling frequency obtained part (ii), find x(n) for this signal. 2

(iv) Calculate the first four samples' values for the obtained x(n). 1.5

Question 2 [8 Points] [LO's: 3]

a- Prove that the DFT, X(K), of a sequence x(n) is periodic. 1

b- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the DFT method, find the frequency components of this sequence, X(k). 2

(ii) Check your results obtained in part (i) by calculating x(n) using the IFFT algorithm. 2

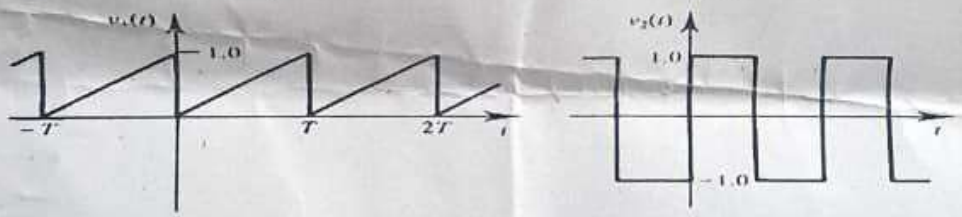
(iii) Using your results, verify Parseval's theorem. 1.5

(iv) Calculate and plot the amplitude and the phase spectrums of X(k). 1.5

Question 3 [8 Points] [LO's: 3]

a- Determine the output of an electrical system of impulse response function h(n) = {0, 0.8, 0.99, 0.9, 1}, when the input is, x(n) = {0, 2.5, 5.0, 0} volts are applied. 3.5

b- Obtain the cross-correlation $r_{12}(-\tau)$ between the waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$ of Fig. 1 4.5



1.5

Fig. 1 The waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$.

Question 4 [8 Points] [LO's: 4]

a- For LTIS define the following items: Linearity, Stability and difference equation. 1.5

b- For the following transfer function, H(z), given by: $H(z) = \frac{z^2(z+0.5)}{(z-0.5)(z-1)^2}$

(i) Plot the pole - zero diagram of H (z). 1.5

(ii) Find the difference equation of this filter, y (n). 1.5

(iii) A digital system has the transfer function, H(z), using the power series method (recursive formula)find the first "4" values for its impulse response, h(n). 3.5

Question 5 [8 Points] [LO's: 2, 4]

a- The transfer function for an FIR filter is given by: $H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$. Draw the realization block diagram for each of the following cases:

1- Transversal structure; 1

2- A two-stage lattice structure. 1.5

Then calculate the values of the coefficients for the lattice structure. 1.5

b- Check (√) for the right statements and (x) for the wrong statements and correct the mistakes. 4

(i) FIR requires more coefficients for sharp cutoff filters than IIR. ( )

(ii) For a given amplitude response specification, more processing time and storage will be required for IIR filter implementation. ( )

(iii) IIR filters are used if little or no phase distortion is desired. ( )

(iv) IIR filters can have an exactly linear phase response. ( )

Good Luck